

Comparison of Tokenization and Embedding Methods on an NER Pipeline

This report summarizes the performance of disease named entity recognition (NER) pipelines developed using the BC5CDR biomedical dataset. The task involved assigning BIO tags (B-Disease, I-Disease and O) to each token in a sentence. The pipeline consisted of tokenization, embedding generation, sequence modeling and token classification stages. Five embedding methods (Word2Vec, GloVe, FastText, ELMo and BERT) were evaluated in combination with three tokenization approaches (Whitespace, NLTK and WordPiece).

Embedding	Tokenizer	F1-Score
Word2Vec	Whitespace	0.51
Word2Vec	WordPiece	0.44
Word2Vec	NLTK	0.50
GloVe	Whitespace	0.62
GloVe	WordPiece	0.63
GloVe	NLTK	0.62
FastText	Whitespace	0.77
FastText	WordPiece	0.79
FastText	NLTK	0.79
ELMo	Whitespace	0.28
ELMo	WordPiece	0.29
ELMo	NLTK	0.29
BERT	Whitespace	0.77
BERT	WordPiece	0.74
BERT	NLTK	0.74

Best-Performing Pipeline

Among all the tested combinations, **FastText** gave the best overall results. It achieved the highest F1 score (**0.79**) when used with both the **NLTK** and **WordPiece** tokenizers. This suggests that FastText is particularly good at recognizing disease names because it can capture information from parts of words which is useful for complex biomedical terms.

Overall the results show that **FastText combined with either NLTK or WordPiece tokenization was the most effective pipeline for disease entity recognition.**